

Work, Energy and Power, solved

Step-by-step solutions to every practice problem in Chapter 5. Try each problem on your own before you read its solution.

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P5.1

Kinetic friction can do positive work. Drop a box onto a moving conveyor belt. Until the box matches the belt's speed, the belt slides under it, and kinetic friction pulls the box **forward**, in the direction of its displacement relative to the ground. So friction does positive work on the box, increasing its kinetic energy.

Static friction can do positive work. A crate on the bed of an accelerating truck is sped up by static friction, which points forward along the crate's displacement. Static friction from the ground on your foot as you start running is another example.

The catch. Work depends on the frame of reference. In the belt's frame, the box moves backward and friction does negative work on it. Also, the *total* work done by a friction pair on the two surfaces together is never positive: kinetic friction always converts some mechanical energy into heat.

P5.2

Positive work: you stand in a lift that is moving upward. The floor's normal force points up, and your displacement is up, so $W_N > 0$. This is how the lift motor's energy reaches you.

No work: a block sliding along a fixed horizontal floor. The normal force is vertical and the displacement horizontal, so $W_N = 0$. In general, the normal force from a fixed surface does no work, because the motion is always along the surface.

P5.3

In a circular orbit, gravity always points towards the centre, perpendicular to the velocity. So at every instant

$$P = \vec{F} \cdot \vec{v} = 0.$$

Gravity therefore does **zero work** over any part of the orbit: a quarter, a half or the whole. That is consistent with the constant speed, since $\Delta K = 0$.

Over a full orbit the work would be zero for *any* closed path, because gravity is conservative. In an *elliptical* orbit, gravity does positive work on the way in and negative work on the way out, so the speed changes (Chapter 8).

P5.4

Final speeds are equal. Only gravity does work, since the normal force is perpendicular to the motion. Energy conservation gives $\frac{1}{2}mv^2 = mgh$, so $v = \sqrt{2gh}$ for both blocks, whatever the slope.

Times differ. On a ramp of angle θ , the acceleration is $g \sin \theta$ and the length is $L = h / \sin \theta$. From $L = \frac{1}{2}at^2$:

$$t = \sqrt{\frac{2L}{g \sin \theta}} = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}.$$

The **gentle, long ramp** takes longer, by the factor $\sin \theta_{\text{steep}} / \sin \theta_{\text{gentle}}$. Energy tells you *how fast*. It does not tell you *how long*.

P5.5

$$W = \int_1^3 (3x^2 - 2x) dx = \left[x^3 - x^2 \right]_1^3 = (27 - 9) - (1 - 1) = 18 \text{ J}.$$

P5.6

(a) **Frictionless.** All the kinetic energy goes into the spring:

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow x = v\sqrt{\frac{m}{k}} = 4.0\sqrt{\frac{2.0}{200}} = 4.0(0.10) = 0.40 \text{ m.}$$

(b) **With friction under the spring.** Friction $\mu_k mg = 0.20(2.0)(9.8) = 3.92 \text{ N}$ acts over the compression distance x :

$$\begin{aligned} \frac{1}{2}mv^2 &= \frac{1}{2}kx^2 + \mu_k mgx \Rightarrow 16 = 100x^2 + 3.92x. \\ x &= \frac{-3.92 + \sqrt{3.92^2 + 4(100)(16)}}{2(100)} = \frac{-3.92 + \sqrt{6415}}{200} = \frac{-3.92 + 80.1}{200} = 0.381 \text{ m.} \end{aligned}$$

Check. Slightly less than 0.40 m, as expected: friction absorbs $3.92 \times 0.381 = 1.5 \text{ J}$ of the original 16 J.

P5.7

Speed. The bob drops through a height $L(1 - \cos 60^\circ) = 2.0(0.5) = 1.0 \text{ m}$:

$$v = \sqrt{2g(1.0)} = \sqrt{19.6} = 4.43 \text{ m/s.}$$

Tension. At the bottom the radial equation gives

$$T = mg + \frac{mv^2}{L} = 0.50(9.8) + \frac{0.50(19.6)}{2.0} = 4.9 + 4.9 = 9.8 \text{ N.}$$

That is exactly $2mg$, a neat result for a release angle of 60° .

P5.8

Force needed at constant speed. The engine must balance the component of gravity along the road plus the resistance. With a 5% grade, $\sin \theta = 5/100 = 0.050$:

$$F = mg \sin \theta + 600 = 1500(9.8)(0.050) + 600 = 735 + 600 = 1335 \text{ N.}$$

Power:

$$P = Fv = 1335(20) = 2.67 \times 10^4 \text{ W} = 26.7 \text{ kW.}$$

Of this, 14.7 kW goes into gaining height and 12 kW into overcoming resistance.

P5.9

$$\Delta E_{\text{lost}} = mgh - \frac{1}{2}mv^2 = 70(9.8)(40) - \frac{1}{2}(70)(22)^2 = 27\,440 - 16\,940 = 10\,500 \text{ J.}$$

As a fraction of the initial potential energy: $10\,500/27\,440 = 0.38$, about 38%.

Without losses, the skier would have reached $\sqrt{2g(40)} = 28 \text{ m/s}$.

P5.10

(a) Volume flow rate:

$$Q = \frac{5.0 \times 10^{-3} \text{ m}^3}{60 \text{ s}} = 8.33 \times 10^{-5} \text{ m}^3/\text{s}.$$

The work per unit time is the pressure times the volume pumped per unit time:

$$P = pQ = (1.3 \times 10^4 \text{ Pa})(8.33 \times 10^{-5} \text{ m}^3/\text{s}) = 1.08 \text{ W}.$$

(Units: $\text{Pa} \times \text{m}^3/\text{s} = \text{N}/\text{m}^2 \times \text{m}^3/\text{s} = \text{N m/s} = \text{W}$.)

(b) Per day:

$$W = 1.08 \times 86\,400 = 9.4 \times 10^4 \text{ J} \approx 94 \text{ kJ}.$$

That is enough to lift the person (70 kg) about 140 m.

(c) Metabolic power: $1.08/0.15 = 7.2 \text{ W}$.

(d) The power is proportional to pQ :

$$\frac{P_{\text{ex}}}{P_{\text{rest}}} = \frac{25}{5.0} \times \frac{16}{13} = 5 \times 1.23 = 6.2.$$

So the output rises to about 6.7 W.

Evaluate. The kinetic energy given to the blood is a small extra (a few percent at rest). The right ventricle adds about another 20% of the left ventricle's work, because pulmonary pressures are lower. This calculation underlies the clinical "stroke work" measures used to assess heart failure.

P5.11

Each second, a mass $\rho Q = 1000(500) = 5.0 \times 10^5$ kg of water falls 100 m, releasing gravitational energy at the rate

$$P_{\text{in}} = \rho Q g h = 5.0 \times 10^5 (9.8) (100) = 4.9 \times 10^8 \text{ W}.$$

Electrical output:

$$P_{\text{out}} = 0.90 \times 4.9 \times 10^8 = 4.4 \times 10^8 \text{ W} = 441 \text{ MW}.$$

That is enough for a city of a few hundred thousand homes.

P5.12

(a) Mechanical energy at the top of each bounce is mgh , so the fraction kept is $6.4/10 = 0.64$. 36% is lost in the bounce, mostly as heat and sound.

(b) Just before the bounce:

$$v_1 = \sqrt{2g(10)} = 14.0 \text{ m/s}.$$

Just after:

$$v_2 = \sqrt{2g(6.4)} = 11.2 \text{ m/s}.$$

The ratio $v_2/v_1 = 0.80$ is the *coefficient of restitution* (Chapter 6).

(c) Each bounce multiplies the height by 0.64:

$$h_4 = 10(0.64)^4 = 10(0.168) = 1.68 \text{ m}.$$

P5.13

(a) **Lowest point.** The jumper falls 20 m freely, then stretches the cord by x . At the lowest point $v = 0$, so all the gravitational energy lost is stored in the cord:

$$mg(20 + x) = \frac{1}{2} kx^2 \Rightarrow 686(20 + x) = 125x^2 \Rightarrow 125x^2 - 686x - 13720 = 0.$$

$$x = \frac{686 + \sqrt{686^2 + 4(125)(13\,720)}}{250} = \frac{686 + \sqrt{7.33 \times 10^6}}{250} = \frac{686 + 2707}{250} = 13.6 \text{ m.}$$

Total fall: $20 + 13.6 = 33.6 \text{ m.}$

(b) **Maximum deceleration** is at the lowest point, where the cord force is largest:

$$F_{\text{net}} = kx - mg = 250(13.57) - 686 = 3393 - 686 = 2707 \text{ N,}$$

$$a = \frac{2707}{70} = 38.7 \text{ m/s}^2 \approx 3.9 g \text{ upward.}$$

(c) **Maximum speed** is where the acceleration is zero, so $kx = mg$:

$$x_0 = \frac{686}{250} = 2.74 \text{ m of stretch, that is, after a fall of } 22.74 \text{ m.}$$

By energy conservation:

$$\frac{1}{2}mv^2 = mg(22.74) - \frac{1}{2}k(2.74)^2 = 15\,600 - 941 = 14\,660 \text{ J} \Rightarrow v = \sqrt{\frac{2(14\,660)}{70}} = 20.5 \text{ m/s.}$$

Evaluate. The speed keeps increasing for a short way after the cord goes taut, until the cord force matches the weight. That is just past the 19.8 m/s reached in the 20 m of free fall. Real bungee cords have k chosen so that the peak load stays below about 3–4 g.

P5.14

Release height $h = 3R$. Use $v^2 = 2g(3R - y)$ at height y .

Bottom ($y = 0$): $v^2 = 6gR$. Radial (up positive):

$$N - mg = \frac{mv^2}{R} = 6mg \Rightarrow N = 7mg.$$

Level with the centre ($y = R$): $v^2 = 4gR$. Here the normal force is horizontal, towards the centre, and gravity is tangential, so it does not contribute to the radial equation:

$$N = \frac{mv^2}{R} = 4mg.$$

Top ($y = 2R$): $v^2 = 2gR$. Both forces point down, towards the centre:

$$N + mg = \frac{mv^2}{R} = 2mg \Rightarrow N = mg.$$

Check. With $h = 2.5R$ we found $N_{\text{top}} = 0$. Raising the release point by $\frac{1}{2}R$ adds mg of normal force at the top. ✓

P5.15

Set up. Let θ be the angle from the vertical through the centre of the hemisphere. The particle has dropped a height $R(1 - \cos \theta)$, so by energy conservation

$$v^2 = 2gR(1 - \cos \theta).$$

Radial equation. The component of gravity towards the centre is $mg \cos \theta$, and the normal force points away from the centre:

$$mg \cos \theta - N = \frac{mv^2}{R}.$$

Leaving the surface. The particle leaves when $N = 0$:

$$g \cos \theta = 2g(1 - \cos \theta) \Rightarrow 3 \cos \theta = 2 \Rightarrow \cos \theta = \frac{2}{3}, \quad \theta = 48.2^\circ.$$

Height above the base: $R \cos \theta = \frac{2}{3}R$.

Speed at that moment: $v^2 = 2gR(1 - \frac{2}{3}) = \frac{2}{3}gR$, so $v = \sqrt{2gR/3}$.

Evaluate. The result does not depend on g , R or m : the particle always leaves one-third of the way down in height. After that it moves as a projectile.

P5.16

(a) **Equilibrium.**

$$\frac{dU}{dr} = -\frac{2A}{r^3} + \frac{B}{r^2} = 0 \Rightarrow r_0 = \frac{2A}{B}.$$

Stability.

$$U''(r) = \frac{6A}{r^4} - \frac{2B}{r^3}.$$

At r_0 , using $r_0^3 = 2Ar_0/B$:

$$U''(r_0) = \frac{1}{r_0^3} \left(\frac{6A}{r_0} - 2B \right) = \frac{1}{r_0^3} (3B - 2B) = \frac{B}{r_0^3} > 0.$$

This is a minimum, so the equilibrium is **stable**.

(b) **Dissociation energy.**

$$U(r_0) = \frac{A}{r_0^2} - \frac{B}{r_0} = \frac{AB^2}{4A^2} - \frac{B^2}{2A} = \frac{B^2}{4A} - \frac{B^2}{2A} = -\frac{B^2}{4A}.$$

Since $U \rightarrow 0$ as $r \rightarrow \infty$, the energy needed to separate the atoms from rest at r_0 is

$$E_{\text{diss}} = 0 - U(r_0) = \frac{B^2}{4A}.$$

(c) **Effective spring constant.** Expanding about the minimum, $U \approx U(r_0) + \frac{1}{2}U''(r_0)(r - r_0)^2$, so

$$k_{\text{eff}} = U''(r_0) = \frac{B}{r_0^3} = \frac{B \cdot B^3}{8A^3} = \frac{B^4}{8A^3}.$$

Evaluate. Every smooth potential looks like a spring near a minimum. That is why molecular vibrations, crystal lattices and many engineering structures oscillate harmonically for small amplitudes (Chapter 9). In infrared spectroscopy, measuring the vibration frequency $\omega = \sqrt{k_{\text{eff}}/\mu}$ reveals bond stiffness.

P5.17

Set up. Take the table top as $U = 0$. The chain has mass per unit length $\lambda = M/L$.

Initial energy. The hanging part has mass λa , with its centre of mass $a/2$ below the table. The part on the table has $U = 0$. So

$$U_i = -\lambda a g \frac{a}{2} = -\frac{Mga^2}{2L}, \quad K_i = 0.$$

Final energy. The whole chain hangs, with its centre of mass $L/2$ below the table:

$$U_f = -\frac{MgL}{2}, \quad K_f = \frac{1}{2}Mv^2.$$

Conservation. The table is frictionless, so mechanical energy is conserved:

$$\frac{1}{2}Mv^2 = \frac{MgL}{2} - \frac{Mga^2}{2L} = \frac{Mg}{2L}(L^2 - a^2) \Rightarrow v = \sqrt{\frac{g(L^2 - a^2)}{L}}.$$

Check. If $a \rightarrow L$ (the chain is already all hanging), $v \rightarrow 0$. ✓ If $a \rightarrow 0$, $v \rightarrow \sqrt{gL}$, but the chain would then need a nudge to start. ✓

P5.18

(a) **Constant power.** With no losses, all the power goes into kinetic energy:

$$\frac{d}{dt} \left(\frac{1}{2} mv^2 \right) = P \Rightarrow \frac{1}{2} mv^2 = Pt \Rightarrow v(t) = \sqrt{\frac{2Pt}{m}}.$$

Integrating:

$$x(t) = \int_0^t \sqrt{\frac{2P}{m}} t'^{1/2} dt' = \frac{2}{3} \sqrt{\frac{2P}{m}} t^{3/2}.$$

(b) **Numbers.**

$$t = \frac{mv^2}{2P} = \frac{1200(27)^2}{2(60\,000)} = \frac{874\,800}{120\,000} = 7.3 \text{ s.}$$

$$x = \frac{2}{3} vt = \frac{2}{3} (27)(7.29) = 131 \text{ m.}$$

(This uses $x = \frac{2}{3} vt$, which follows from the two formulas in part (a).)

(c) **What limits the start.** With constant power, $a = P/(mv)$, which is infinite at $v = 0$. In reality:

1. The tyres can push the car forward only through static friction, so $a \leq \mu_s g \approx 8\text{--}10 \text{ m/s}^2$ on a dry road. The start is *traction-limited*, which is why launch control systems manage wheel slip.
2. An internal-combustion engine cannot deliver full power at low revs, so the start is also *torque-limited*. Gearing exists to manage this.
3. Electric motors deliver maximum torque from zero speed, which is why electric cars accelerate so well from rest, until traction limits them.

The constant-power model becomes accurate only once the car is moving fast enough that $P/(mv) < \mu_s g$.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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